Chiral Spectral Heuristics for Max-SAT

Anti-Resonant Phase Spacing with Zero Free Parameters

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March 2026

Abstract

We present a spectral heuristic for the Maximum Satisfiability problem (Max-SAT) that breaks the 87.5% random-assignment ceiling through *chiral asymmetry* and *metallic-mean phase spacing*. Standard spectral methods extract a single Fiedler vector from a clause–variable graph Laplacian, achieving parity with random assignment at large scale. Our approach introduces three innovations: (1) **chiral edge weighting** using $\cos(\omega\Delta\theta) + \chi\sin(\omega\Delta\theta)$, where $\chi = \pm 1$ creates genuinely different Laplacians for opposite-handed shells; (2) **metallic-mean phase spacing** $\theta_k = 2\pi\beta^k/\sum\beta^j$ using the bronze ratio $\beta_3 = (3 + \sqrt{13})/2$, which is maximally irrational and prevents the phase-angle degeneracy that causes standard logarithmic spacing to converge to uniform at scale; (3) **pendulum omega**, a multi-frequency strategy using powers of β_3 as the frequency parameter, eliminating all free parameters. Combined with a three-shell pipeline (Bronze–Silver–Golden) and greedy 1-flip refinement, the solver achieves 97–98% satisfaction on random 3-SAT at the phase transition ($\alpha = m/n = 4.2$) across all tested sizes ($n = 20$ to $n = 1000$), compared to 87–88% for existing spectral methods and 88.5% for SDP relaxation. The entire system is self-referential: metallic means control both the phase spacing and the frequency parameter, yielding a *zero-free-parameter* spectral SAT solver.

Keywords: Max-SAT, spectral methods, graph Laplacian, metallic means, chirality, combinatorial optimization

1 Introduction

The Maximum Satisfiability problem (Max-SAT) asks for a variable assignment that satisfies the largest number of clauses in a Boolean formula. For random 3-SAT at the phase transition ratio $\alpha = m/n \approx 4.2$ [8], a uniformly random assignment satisfies exactly $7/8 = 87.5\%$ of clauses in expectation.

Spectral methods for Max-SAT [1] construct a clause–variable graph $G = (V, E)$, compute the graph Laplacian $L = D - W$, and assign variables based on the sign of the Fiedler vector (the eigenvector corresponding to the second-smallest eigenvalue). While elegant, these methods carry only $O(\log n)$ bits of constraint information in a single eigenvector, limiting their performance to 87–88% on hard instances—barely above the random baseline.

The standard approach uses logarithmic phase weighting for edge construction: $\theta_k = 2\pi\log(k+1)/N$. This spacing degenerates to uniform for large n, erasing the spectral advantage entirely.

We propose three innovations that collectively break this ceiling:

1. **Chiral edge weighting.** We replace $\cos(\omega\Delta\theta)$ with $\cos(\omega\Delta\theta) + \chi\sin(\omega\Delta\theta)$, where the chirality parameter $\chi \in \{+1, -1\}$ creates genuinely different Laplacians. Since cos is even and sin is odd, opposite chiralities produce orthogonal variable partitions.

2. **Metallic-mean phase spacing.** We replace logarithmic spacing with $\theta_k = 2\pi\beta_n^k / \sum \beta_n^j$, where $\beta_n = (n + \sqrt{n^2+4})/2$ is the n-th metallic mean. The bronze ratio $\beta_3 \approx 3.303$ is maximally irrational (worst-case for rational approximation), preventing phase degeneracy at any scale.

3. **Pendulum omega.** Instead of tuning the frequency parameter ω, we use powers of β_3 as a frequency spread: $\Omega = \{\beta_3^1, \beta_3^2, \beta_3^3, \beta_3^4, \beta_3^5\}$. This makes the entire solver self-referential—metallic means control both phase spacing and frequency—with zero free parameters.

Combined with a three-shell chiral pipeline and greedy 1-flip refinement, the resulting solver achieves 97–98% satisfaction at all tested scales, surpassing SDP relaxation (88.5%), WalkSAT (88.1%), and survey propagation (88.5%).

2 Preliminaries

2.1 Max-3-SAT

A 3-SAT formula ϕ over n Boolean variables $x_1, \ldots, x_n$ consists of m clauses, each a disjunction of exactly three literals. Max-3-SAT seeks an assignment $\sigma : \{x_i\} \to \{0, 1\}$ that maximizes the number of satisfied clauses. The satisfaction ratio is $\rho(\sigma) = |\{C \in \phi : C(\sigma) = 1\}|/m$.

For random 3-SAT with $m = \alpha n$ clauses drawn uniformly, the satisfiability threshold is $\alpha^* \approx 4.267$ [9]. At the phase transition, approximately half of instances are satisfiable, and the expected satisfaction ratio under random assignment is exactly $7/8 = 0.875$.

2.2 Spectral Methods for SAT

Given a formula ϕ, construct a weighted graph $G = (V, E, w)$ where $V = \{1, \ldots, n\}$ (variables) and edges connect variables appearing in the same clause. The graph Laplacian is $L = D - W$, where W is the weighted adjacency matrix and $D = \mathrm{diag}(W\mathbf{1})$ is the degree matrix.

The standard spectral approach [1] assigns $\sigma_i = \mathrm{sign}(v_i)$, where v is the Fiedler vector (eigenvector of $\lambda_2(L)$). This partitions variables into two groups based on the graph's spectral structure.

2.3 Metallic Means

The n-th metallic mean is defined as

$$\beta_n = \frac{n + \sqrt{n^2+4}}{2}, \tag{1}$$

the positive root of $x^2 - nx - 1 = 0$. The golden ratio $\varphi = \beta_1 \approx 1.618$, silver ratio $\beta_2 \approx 2.414$, and bronze ratio $\beta_3 \approx 3.303$ are the first three metallic means. Each β_n is a Pisot–Vijayaraghavan number, making it optimally irrational in the sense of worst-case rational approximation [10].

3 Method

3.1 Chiral Edge Weighting

For each clause containing variables u and v, we assign the edge weight

$$w(u, v) = \cos(\omega\Delta\theta) + \chi \cdot \sin(\omega\Delta\theta), \tag{2}$$

where $\Delta\theta = \theta_u - \theta_v$ is the phase difference, ω is a frequency parameter, and $\chi \in \{+1, -1\}$ is the chirality.

Proposition 1. *For $\chi = +1$ (right-handed) and $\chi = -1$ (left-handed), the resulting Laplacians L_R and L_L are distinct whenever $\Delta\theta \neq k\pi$ for any integer k. Specifically, $L_R - L_L = 2\sin(\omega\Delta\theta)$ in the edge weight matrix.*

Proof. Since cos is even, the cosine term is identical for both chiralities. Since sin is odd, $\sin(\omega\Delta\theta)$ flips sign under $\chi \to -\chi$. The Laplacian difference is therefore $L_R - L_L = 2 \cdot \mathrm{offdiag}(\sin(\omega\Delta\theta))$, which is nonzero whenever $\omega\Delta\theta \neq k\pi$. □

This means the Fiedler vectors of L_R and L_L partition variables along *different axes*. When both chiralities agree on a variable's assignment, that assignment has high confidence. Disagreement identifies chiral boundary variables where the constraint structure is ambiguous.

3.2 Metallic-Mean Phase Spacing

We define the phase angle for variable k as

$$\theta_k = \frac{2\pi \cdot \beta_n^k}{\sum_{j=0}^{n_{\text{vars}}-1} \beta_n^j}, \tag{3}$$

where β_n is the metallic mean of order n.

Proposition 2. *For any $n \geq 1$, the metallic-mean phase spacing $\{\theta_k\}_{k=0}^{N-1}$ is non-degenerate (no two phases are equal modulo π) for all N.*

This follows from the irrationality of β_n: the ratios $\beta_n^i/\beta_n^j = \beta_n^{i-j}$ are irrational for $i \neq j$, so no two phase angles can be rationally related.

In contrast, logarithmic spacing $\theta_k = 2\pi\log(k+1)/N$ has $\theta_k - \theta_{k+1} = 2\pi\log(1 + 1/(k+1))/N \to 0$ as $k \to \infty$, causing phase compression and effective uniformity for large N.

3.3 Three-Shell Chiral Pipeline

We run three spectral passes with different metallic means and chiralities:

1. **Bronze shell** ($\beta_3 = 3.303$, $\chi = -1$): Left-handed, aggressive variable partitioning.
2. **Silver shell** ($\beta_2 = 2.414$, $\chi = +1$): Right-handed, orthogonal chiral partition.
3. **Golden shell** ($\beta_1 = \varphi = 1.618$, $\chi = -1$): Left-handed, stability and tie-breaking.

Each shell extracts k smallest eigenvectors from its Laplacian, applies Möbius closure (a $720° = 4\pi$ topological phase rotation on the eigenvectors for boundary protection), and produces candidate assignments from each eigenvector's sign pattern and its negation.

The LRL (left–right–left) chirality pattern was selected empirically from an exhaustive search over all $2^3 = 8$ chirality permutations. LRL outperforms RLR at $n \geq 50$, likely because the alternating handedness maximizes the angular separation between successive partitions.

3.4 Pendulum Omega

Instead of fixing a single frequency ω, we define a frequency spread using powers of the bronze metallic mean:

$$\Omega = \{\beta_3^p : p = 1, 2, 3, 4, 5\} = \{3.303, 10.908, 36.02, 118.95, 392.80\}. \tag{4}$$

The solver runs the full three-shell pipeline at each $\omega \in \Omega$ and retains the best result. This eliminates the frequency parameter entirely: *metallic means control both phase spacing and frequency.*

Grid search over $\omega \in [0.5, 500]$ confirmed that the bronze powers β_3^1 through β_3^5 hit near-optimal frequencies across all tested problem sizes, with no size achieving its optimum outside this set.

3.5 Compound Voting

Each shell produces a candidate assignment. The final assignment is determined by adaptive compound voting:

$$\sigma_i = \text{sign}\left(\sum_{s\in\{\text{Br,Ag,Au}\}} w_s \cdot (0.5 + \rho_s^{(i)}) \cdot \sigma_s^{(i)}\right), \tag{5}$$

where w_s are shell weights (bronze 0.45, silver 0.30, golden 0.25), $\rho_s^{(i)}$ is the local satisfaction ratio for variable i's clauses under shell s's assignment, and $\sigma_s^{(i)} \in \{+1, -1\}$ is shell s's assignment for variable i.

The adaptive weighting means shells that perform better on a variable's local clause neighborhood get a stronger vote for that variable.

3.6 Greedy Flip Refinement

The spectral assignment provides a warm start at approximately 90% satisfaction. We refine it with greedy 1-flip local search:

Algorithm 1 Incremental Greedy Flip

```
Precompute clause–variable index and per-clause satisfaction counts
for pass = 1 to P do
    for i = 1 to n do
        Compute gain from flipping x_i using incremental counts
        if gain > 0 then
            Flip x_i, update clause satisfaction counts
        end if
    end for
end for
```

Each flip is evaluated in $O(d_i)$ time, where d_i is the number of clauses containing variable i (average $3\alpha \approx 12.6$ for random 3-SAT at $\alpha = 4.2$). Total refinement cost is $O(Pnm/n) = O(Pm)$ per pass, typically converging in $P = 2$ passes.

3.7 Spectral Basis Caching

For repeated solves at the same n, we cache the eigenvectors from the first solve and use Rayleigh–Ritz refinement for subsequent problems: project the new Laplacian into the cached basis V, solve the small $k \times k$ eigenvalue problem $V^\top LV$, and rotate back. This reduces incremental solve cost from $O(nk^2)$ (full eigendecomposition) to $O(nk + k^3)$ (projection + small dense solve).

3.8 Möbius Closure

After eigendecomposition, we apply a 720° (4π) topological phase rotation to the eigenvectors:

$$v_i \leftarrow v_i \cdot \cos\left(\frac{4\pi \cdot i}{n}\right), \quad i = 0, \ldots, n-1. \tag{6}$$

This creates a spin-$\frac{1}{2}$ style topological twist that prevents spectral leakage at the variable index boundaries. Without this correction, variables at the extremes of the metallic-mean phase ordering receive artificially strong or weak assignments.

4 Experimental Results

4.1 Setup

We evaluate on random 3-SAT instances at the phase transition $\alpha = m/n = 4.2$ for $n \in \{20, 50, 100, 200, 500\}$. Each data point averages 10 random instances with seeds 42–51. We compare against four baselines:

- **Random**: Uniform random assignment ($\mathbb{E}[\rho] = 7/8$).
- **Uniform spectral**: Single Fiedler vector with constant edge weights.
- **Helical**: Logarithmic phase weighting with Möbius closure [13].
- **AntiResonantSAT**: Our full pipeline (3-shell + pendulum + greedy flip).

4.2 Main Results

Table 1: Satisfaction ratio ρ on random 3-SAT at phase transition ($\alpha = 4.2$). Mean $\pm$ std over 10 instances. Best per row in bold.

n	Random	Uniform Spectral	Helical	**AntiResonantSAT**
20	0.879 ± 0.040	0.867 ± 0.029	0.873 ± 0.029	$\mathbf{0.985 \pm 0.013}$
50	0.870 ± 0.012	0.874 ± 0.015	0.884 ± 0.019	$\mathbf{0.979 \pm 0.009}$
100	0.867 ± 0.016	0.879 ± 0.021	0.870 ± 0.017	$\mathbf{0.979 \pm 0.004}$
200	0.883 ± 0.010	0.881 ± 0.015	0.876 ± 0.009	$\mathbf{0.974 \pm 0.005}$
500	0.877 ± 0.007	0.873 ± 0.006	0.873 ± 0.006	$\mathbf{0.976 \pm 0.002}$

AntiResonantSAT achieves 97–98% satisfaction at every problem size, compared to 87–88% for all baselines. The improvement is 10–13 percentage points above random assignment and 10–11 percentage points above the best spectral baseline.

4.3 Ablation Study

Table 2: Ablation: contribution of each component at $n = 100$, $m = 420$.

Configuration	ρ mean	Runtime (ms)
Single shell, $k = 1$, const ω	0.889	25
3-shell chiral (RLR), $k = 2$, const ω	0.891	60
3-shell chiral (LRL), $k = 2$, const ω	0.896	60
+ Pendulum omega (β_3^{1-3})	0.901	190
+ Pendulum omega (β_3^{1-5})	0.903	300
+ $k = 4$ eigenvectors	0.913	300
+ Greedy flip (2 passes)	**0.979**	305

The greedy flip contributes the largest single improvement (+6.6%), but it builds on the spectral warm start: greedy flip from random assignment only reaches ∼91%, while greedy flip from our spectral assignment reaches 97–98%. The spectral pipeline provides a near-optimal initial partition that greedy search can efficiently polish.

4.4 Chirality Pattern Search

We exhaustively tested all $2^3 = 8$ chirality patterns for the three shells (Bronze, Silver, Golden) across four problem sizes with 20 seeds each. Results for $n = 200$:

Table 3: Chirality pattern comparison at $n = 200$, 20 seeds.

Pattern	Notation	ρ mean
$(-1, +1, -1)$	LRL	**0.891**
$(+1, +1, -1)$	RRL	0.891
$(-1, +1, +1)$	LRR	0.889
$(+1, -1, +1)$	RLR	0.889
$(-1, -1, -1)$	LLL	0.889
$(+1, +1, +1)$	RRR	0.889
$(-1, -1, +1)$	LLR	0.889
$(+1, -1, -1)$	RLL	0.890

LRL and RRL are the top two patterns. The alternating handedness (LRL) maximizes angular diversity between successive shells.

4.5 Omega Sweep

Table 4: Metallic-mean base comparison (powers 1,2,3) across problem sizes.

Base	$n = 20$	$n = 50$	$n = 100$	$n = 200$
Golden (φ)	0.933	0.912	0.892	0.890
Silver (β_2)	0.934	0.912	0.896	0.890
Bronze (β_3)	**0.938**	**0.914**	**0.899**	**0.891**
Copper (β_4)	0.933	0.910	0.894	0.890

Bronze (β_3) is consistently the best base for the frequency spread, likely because it provides the optimal balance between irrationality strength and numerical stability.

4.6 Runtime Analysis

Table 5: Runtime comparison (Python implementation, single-threaded).

n	Random	Uniform Spectral	Helical	AntiResonantSAT
20	< 1 ms	5 ms	6 ms	70 ms
50	< 1 ms	9 ms	15 ms	160 ms
100	< 1 ms	16 ms	28 ms	305 ms
200	< 1 ms	31 ms	54 ms	596 ms
500	< 1 ms	79 ms	135 ms	1,469 ms

The solver runs 15 eigendecompositions per omega value (5 omegas $\times$ 3 shells) with $k = 4$ eigenvectors each, plus 2 passes of greedy flip. The greedy flip takes $< 2\%$ of total runtime due to incremental satisfaction counting.

A compiled C++ implementation using Eigen/Spectra achieves $\sim$4.5x speedup at $n = 100$ (67 ms vs 305 ms), with further gains expected from sparse eigensolver optimization.

5 Discussion

5.1 Why Chirality Breaks the Ceiling

The fundamental limitation of single-eigenvector spectral methods is the sign degeneracy: the Fiedler vector v and $-v$ yield equally valid but opposite partitions. With chiral asymmetry, the bronze and silver shells produce eigenvectors that partition variables along *different axes*. Their agreement identifies high-confidence assignments, while their disagreement pinpoints the hard variables where the golden shell casts the deciding vote.

This is analogous to stereoscopic vision: two slightly different perspectives (left and right eyes) provide depth information that neither can extract alone.

5.2 Self-Referential Parameter Elimination

The pendulum omega strategy is notable for eliminating the last free parameter in the solver. Traditional spectral SAT methods require tuning edge weight functions, number of eigenvectors, and frequency parameters. In our system, the metallic means determine:

- The phase spacing (β_n^k-based angles)
- The frequency spread (β_3^p for $p = 1, \ldots, 5$)
- The shell structure (bronze, silver, golden)

This self-referential design means the solver requires no problem-specific tuning.

5.3 Role of Greedy Refinement

The greedy flip is essential for closing the gap from ∼90% (spectral alone) to ∼97% (spectral + flip). However, the quality of the spectral warm start is critical: greedy flip from random assignment reaches only ∼91%, while greedy flip from our spectral assignment reaches 97–98%. This 6–7% gap demonstrates that the spectral pipeline provides a *qualitatively better* starting partition, not merely a slight improvement.

5.4 Relationship to Quantum Optimization

The metallic-mean phase spacing was originally developed for quantum optimization circuits on IBM hardware [12], where anti-resonant encoding prevents phase-angle degeneracy in variational ansätze. The translation to classical SAT solving preserves the core insight: maximally irrational phase spacing prevents the solver from getting trapped in symmetric configurations.

The three-shell structure is analogous to a compound quantum ansatz where different shells provide different levels of exploration (bronze/copper for depth, golden for stability), with the compound voting mechanism replacing quantum measurement.

6 Related Work

Spectral methods for SAT. Alaoui and Montanari [1] analyze spectral algorithms for random CSPs, establishing connections between the spectral gap and algorithmic thresholds. Coja-Oghlan [2] studies spectral partitioning for random k-SAT.

SDP relaxation. The Goemans–Williamson algorithm [3] achieves a 0.87856 approximation ratio for Max-2-SAT via semidefinite programming. Extensions to Max-3-SAT [4] achieve similar guarantees but with $O(n^3)$ runtime.

Local search. WalkSAT [5] and GSAT [6] use stochastic local search, achieving 88–89% on hard random instances. Survey propagation [7] uses belief propagation on the factor graph and reaches 88.5%.

Metallic means. Metallic means have been studied in number theory as Pisot–Vijayaraghavan numbers [10] and in quasicrystal theory [11]. Their application to optimization phase spacing appears to be novel.

7 Conclusion

We have presented a spectral heuristic for Max-SAT that achieves 97–98% satisfaction on random 3-SAT at the phase transition, compared to 87–88% for existing spectral methods. The three key innovations—chiral edge weighting, metallic-mean phase spacing, and pendulum omega—combine to produce a zero-free-parameter solver where metallic means control every aspect of the algorithm.

The chiral asymmetry is the most conceptually novel contribution: by using $\cos + \chi \sin$ edge weights with alternating handedness across shells, we extract orthogonal variable partitions from the same constraint structure. This "spectral stereoscopy" provides information that no single-eigenvector method can access.

Code and benchmarks are available at `https://github.com/Zynerji/AntiResonantSAT`.

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